

MCKINNEY, TX, USA

WMO: 722541

Lat: 33.190N

Lon: 96.591W

Elev: 586

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 53914

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	25.2	9.1	8.9	27.7	13.7	11.3	33.6	26.5	52.2	24.5	53.9	7.1	0	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.7	100.5	74.6	98.7	75.0	96.4	75.0	78.8	91.0	78.0	90.5	77.2	89.6	9.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.7	137.4	82.8	74.9	133.5	82.2	73.8	128.6	81.6	42.7	91.5	41.7	91.0	40.9	89.7	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.9	20.5	18.5	DB	14.1	104.0	3.9	3.5	11.4	106.5	9.1	108.6	6.9	110.5	4.1	113.1	(4)
			WB	12.7	80.5	3.4	1.8	10.3	81.8	8.3	82.9	6.4	83.9	4.0	85.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	44.7	48.7	56.7	64.4	72.5	80.6	84.2	84.5	77.3	66.3	55.6	46.3	(6)
		DBStd	16.14	9.76	10.57	9.22	7.59	6.45	4.31	3.96	4.49	6.32	8.35	9.38	9.63	(7)
		HDD50	648	221	140	40	4	0	0	0	0	3	50	190		(8)
		HDD65	2471	631	464	283	100	17	0	0	3	86	303	584		(9)
		CDD50	6210	57	102	248	436	699	918	1062	1069	819	507	219	74	(10)
		CDD65	2559	2	6	26	83	250	468	597	604	372	125	22	4	(11)
		CDH74	29375	10	36	165	613	2253	5266	7716	7908	4071	1178	141	18	(12)
		CDH80	14577	0	3	18	108	762	2538	4289	4510	1956	373	19	1	(13)
(14)	Wind	WSAvg	8.1	8.5	9.1	9.7	10.0	8.8	8.1	7.0	6.4	6.3	7.3	8.2	8.3	(14)
(15)	Precipitation	PrecAvg	41.0	2.8	3.0	3.7	3.7	4.8	3.6	2.2	2.4	3.2	4.6	3.3	3.3	(15)
(16)		PrecMax	54.1	6.1	8.7	8.8	8.5	8.1	11.1	6.7	5.9	7.9	10.3	8.2	9.1	(16)
(17)		PrecMin	22.8	0.0	0.5	0.0	1.1	1.8	0.5	0.0	0.0	0.5	0.4	0.5	0.4	(17)
(18)		PrecStd	8.7	1.7	2.2	2.1	2.0	2.1	2.8	1.7	1.6	1.9	3.0	2.3	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	75.2	78.9	82.4	87.4	93.3	98.8	103.0	105.0	101.0	91.2	82.3	76.6		(19)
(20)		0.4% MCWB	60.2	62.4	65.2	70.6	74.5	75.7	73.8	73.5	72.9	70.9	66.4	64.1		(20)
(21)		2% DB	71.1	73.5	78.7	82.5	90.2	96.0	100.0	101.5	96.5	88.0	78.4	71.4		(21)
(22)		2% MCWB	59.7	61.1	64.7	68.6	73.4	75.0	75.2	74.2	73.0	70.8	65.2	61.5		(22)
(23)		5% DB	66.4	70.3	75.0	80.5	87.4	93.3	97.9	99.3	92.9	84.2	74.4	67.3		(23)
(24)		5% MCWB	57.4	60.8	63.1	67.6	72.7	74.9	75.4	75.0	72.9	68.7	63.8	59.8		(24)
(25)		10% DB	62.9	66.3	71.9	77.6	84.2	91.2	95.4	96.9	90.3	81.0	71.1	63.3		(25)
(26)		10% MCWB	54.7	58.4	61.6	65.9	71.5	74.9	75.6	75.1	72.6	67.6	62.5	56.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	66.0	67.9	69.5	74.3	77.5	80.1	80.2	79.6	78.1	76.0	71.1	67.4		(27)
(28)		0.4% MCDB	70.8	73.3	77.4	81.9	88.1	91.5	92.1	91.8	89.1	83.6	76.9	72.0		(28)
(29)		2% WB	63.6	65.8	67.5	71.8	76.0	78.4	78.7	78.4	76.8	74.2	68.4	64.8		(29)
(30)		2% MCDB	67.0	70.7	74.4	79.4	85.5	89.8	91.2	91.6	87.5	82.1	73.9	69.3		(30)
(31)		5% WB	59.8	63.2	65.9	70.3	74.9	77.4	77.9	77.6	75.8	72.6	66.3	61.6		(31)
(32)		5% MCDB	65.0	67.7	72.6	77.2	83.7	88.7	90.6	91.3	86.2	80.3	72.3	65.5		(32)
(33)		10% WB	54.9	59.8	63.9	68.7	73.6	76.4	77.1	76.8	74.8	70.8	63.9	57.3		(33)
(34)		10% MCDB	60.9	65.5	70.2	75.0	81.8	87.2	89.8	90.1	85.0	77.7	69.4	62.2		(34)
(35)	Mean Daily Temperature Range	MDBR	21.5	20.8	20.9	20.7	19.7	19.3	20.7	21.7	21.9	22.4	21.5	20.5		(35)
(36)		5% DB MCDBR	28.4	25.2	23.3	22.3	22.0	21.6	23.7	25.3	24.4	25.0	24.2	25.6		(36)
(37)		5% DB MCWBR	18.2	16.3	13.2	12.0	9.0	6.6	6.1	6.6	7.6	10.7	13.5	16.8		(37)
(38)		5% WB MCDBR	23.0	21.2	19.4	17.8	18.4	18.6	19.9	20.7	19.1	18.9	19.3	20.6		(38)
(39)		5% WB MCWBR	17.9	16.1	13.0	11.6	9.3	6.9	6.6	6.2	6.8	9.6	13.8	16.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.327	0.358	0.394	0.428	0.440	0.443	0.446	0.400	0.357	0.337	0.318		(40)
(41)		taud	2.494	2.469	2.402	2.322	2.289	2.287	2.316	2.294	2.370	2.475	2.477	2.488		(41)
(42)		Ebn at Noon	287	294	292	284	274	269	267	266	274	279	276	278		(42)
(43)		Edh at Noon	27	31	36	41	42	42	41	41	36	30	28	26		(43)
(44)	All-Sky Solar Radiation	RadAvg	888	1041	1384	1685	1822	2057	2099	1931	1621	1270	949	781		(44)
(45)		RadStd	89	158	108	126	167	205	135	126	148	187	121	98		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page